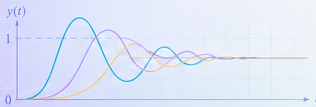
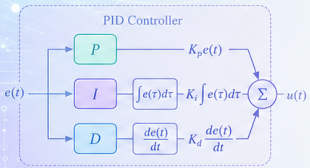
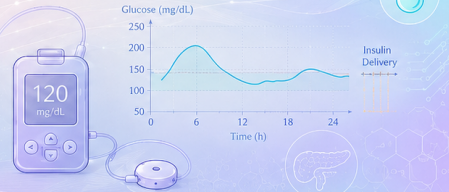
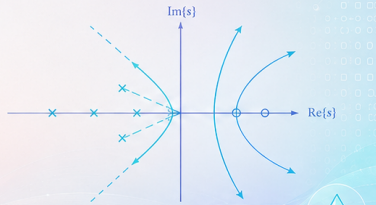
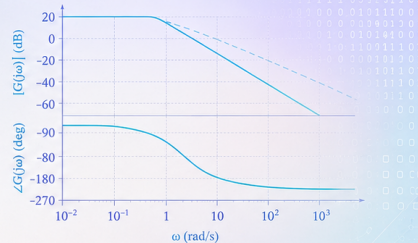


Nyquist Stability

SAHAJ SAXENA
sahajsaxena.github.io



$$G(s) = \frac{Y(s)}{U(s)} \quad a_n \frac{d^n y(t)}{dt^n} + \dots + a_1 \frac{dy(t)}{dt} + a_0 y(t) = b_m \dots + b_m \frac{d^m u(t)}{dt^m} + \dots + b_1 \frac{du(t)}{dt} + b_0 u(t)$$



From Bode Plot to Nyquist Stability

The loop frequency response is

$$L(j\omega) = G(j\omega)H(j\omega).$$

Bode plots display $|L(j\omega)|$ and $\angle L(j\omega)$ separately.

Nyquist Idea

Plot the complex quantity itself:

$$L(j\omega) = \operatorname{Re} L + j \operatorname{Im} L.$$

- ▶ Combines **magnitude and phase** in one complex-plane plot.
- ▶ Infers **closed-loop stability** from the open-loop response.
- ▶ Uses the critical point

$$\boxed{-1 + j0}.$$

Goal: determine stability without explicitly solving the closed-loop characteristic polynomial.

Closed-Loop Stability and the Critical Point

For a standard negative-feedback system,

$$T(s) = \frac{G(s)}{1 + G(s)H(s)}.$$

Define

$$L(s) = G(s)H(s).$$

The closed-loop characteristic equation is

$$1 + L(s) = 0.$$

Therefore,

$$L(s) = -1.$$

Why is $-1 + j0$ Special?

Let

$$F(s) = 1 + L(s).$$

A zero of $F(s)$ is a closed-loop pole.

Since

$$F(s) = 0 \iff L(s) = -1,$$

the point $-1 + j0$ becomes the **critical point** in the Nyquist plot.

10.3 Introduction to the Nyquist Criterion: Contour Mapping

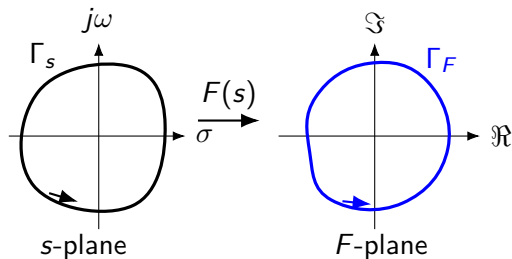
Consider a complex function

$$w = F(s).$$

As s traverses a closed contour Γ_s , the image $F(s)$ traces another closed contour Γ_F .

Central Question

What does the mapped contour reveal about the **zeros and poles** of $F(s)$ enclosed by Γ_s ?



Argument Principle

Let Γ_s enclose

- ▶ P poles of $F(s)$, and
- ▶ Z zeros of $F(s)$,

with no pole or zero lying on the contour.

For the **clockwise Nyquist contour** used here, define N as the net number of **counterclockwise** encirclements of the origin by $F(s)$.

Then

$$\boxed{N = P - Z} \quad \Longleftrightarrow \quad \boxed{Z = P - N}.$$

Interpretation

The change in angle of the mapped contour tells us the difference between the number of poles and zeros of $F(s)$ enclosed by the original contour.

From the Argument Principle to the Nyquist Criterion

Choose

$$F(s) = 1 + L(s), \quad L(s) = G(s)H(s).$$

Poles of $F(s)$ are the poles of $L(s)$. Thus,

Encircling the origin by $F(s) = 1 + L(s)$ is equivalent to encircling

P = number of open-loop RHP poles.

Zeros of $F(s)$ satisfy

$$\boxed{-1 + j0}$$

$$1 + L(s) = 0,$$

by the Nyquist plot of $L(s)$.

Hence,

$$\boxed{Z = P - N},$$

so

where N is the net number of CCW encirclements of -1 .

Z = number of closed-loop RHP poles.

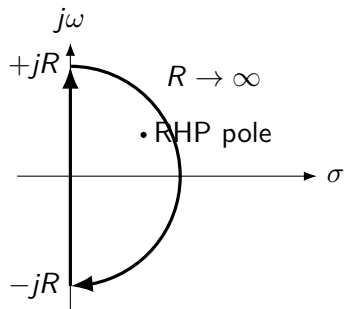
The Nyquist Contour

The contour encloses the entire
right-half s -plane.

- ▶ Traverse the imaginary axis from $-jR$ to $+jR$.
- ▶ Return on a large RHP semicircle.
- ▶ Let

$$R \rightarrow \infty.$$

- ▶ Overall traversal is clockwise.



Nyquist Stability Criterion: Definitions

With the sign convention used in these slides,

$$Z = P - N.$$

Symbol	Meaning
--------	---------

P	Number of poles of $L(s) = G(s)H(s)$ in the open RHP.
N	Net number of counterclockwise encirclements of $-1 + j0$ by the complete Nyquist plot.
Z	Number of closed-loop poles in the open RHP.

Sign Convention

A CCW encirclement contributes $+1$; a clockwise encirclement contributes -1 .

Nyquist Stability Criterion: Stability Condition

Closed-loop stability requires

$$\boxed{Z = 0}.$$

Since

$$Z = P - N,$$

we require

$$\boxed{N = P}.$$

Open-Loop Stable: $P = 0$

The stable closed-loop case requires

$$\boxed{N = 0},$$

i.e. no net encirclement of -1 .

Open-Loop Unstable: $P > 0$

The Nyquist plot must make exactly P net CCW encirclements of the critical point.

Marginal Stability in the Nyquist Diagram

If the Nyquist locus passes through

$$\boxed{-1 + j0},$$

then, for some frequency ω_c ,

$$L(j\omega_c) = -1.$$

Therefore,

$$1 + L(j\omega_c) = 0,$$

and the closed-loop characteristic equation has roots at

$$\boxed{s = \pm j\omega_c}.$$

Meaning

The system is on the **boundary of stability**; sustained oscillation is possible at ω_c .

10.4 Sketching the Nyquist Diagram

To sketch the complete Nyquist diagram of $L(s)$:

1. Evaluate the positive-frequency branch:

$$s = j\omega, \quad 0 \leq \omega < \infty.$$

2. Determine the image of the large semicircle as $|s| \rightarrow \infty$.
3. Use real-coefficient symmetry:

$$\boxed{L(-j\omega) = L^*(j\omega)}.$$

4. Preserve the direction of traversal.
5. Mark the critical point $-1 + j0$.

Practical Shortcut

For real-coefficient transfer functions, sketch $0 \leq \omega < \infty$ first and reflect it about the real axis.

Useful Limits for Sketching

Starting point

$$\lim_{\omega \rightarrow 0^+} L(j\omega).$$

Real-axis crossings

$$\operatorname{Im} L(j\omega) = 0.$$

The corresponding real part locates the crossing.

Always Check

Starting point, endpoint, real-axis crossing, imaginary-axis crossing, symmetry, and direction.

High-frequency endpoint

$$\lim_{\omega \rightarrow \infty} L(j\omega).$$

For a strictly proper transfer function,

$$|L(j\omega)| \rightarrow 0.$$

Imaginary-axis crossings

$$\operatorname{Re} L(j\omega) = 0.$$

Worked Example 1: Sketch a Nyquist Diagram

Consider

$$L(s) = \frac{1}{(s+1)(s+2)}.$$

For $s = j\omega$,

$$L(j\omega) = \frac{2 - \omega^2 - j3\omega}{(2 - \omega^2)^2 + 9\omega^2}.$$

Thus,

$$\operatorname{Re} L = \frac{2 - \omega^2}{(2 - \omega^2)^2 + 9\omega^2}, \quad \operatorname{Im} L = \frac{-3\omega}{(2 - \omega^2)^2 + 9\omega^2}.$$

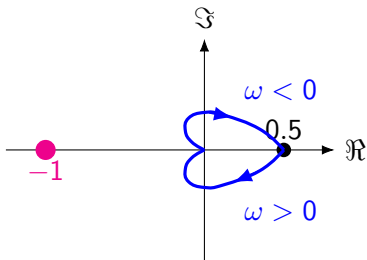
Key points:

$$L(0) = \frac{1}{2},$$

$$\operatorname{Re} L = 0 \text{ at } \omega = \sqrt{2},$$

$$L(j\omega) \rightarrow 0 \text{ as } \omega \rightarrow \infty.$$

Worked Example 1: Nyquist Plot and Stability



Open-loop poles:

$$-1, -2 \Rightarrow P = 0.$$

No encirclement of -1 :

$$N = 0.$$

Therefore,

$$Z = P - N = 0.$$

Conclusion

Closed-loop stable.

Open-Loop Poles on the Nyquist Contour

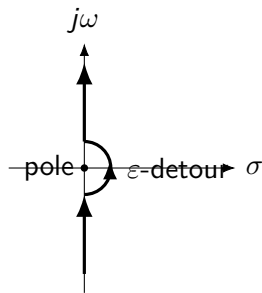
If $L(s)$ has a pole on the imaginary axis, the contour cannot pass through that pole.

A small semicircular **indentation** is used.

For a pole at the origin,

$$s = \varepsilon e^{j\theta}, \quad \varepsilon \rightarrow 0^+.$$

The indentation must also be mapped through $L(s)$.



Example with a Pole on the Contour

Consider

$$L(s) = \frac{s+2}{s^2}.$$

For $s = j\omega$,

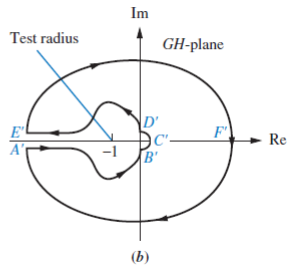
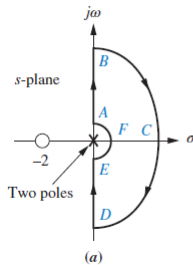
$$L(j\omega) = -\frac{2}{\omega^2} - j\frac{1}{\omega}.$$

Therefore,

$\omega \rightarrow 0^+ : L(j\omega) \rightarrow \infty$ in the third quadrant,

$\omega \rightarrow \infty : L(j\omega) \rightarrow 0$ with phase -90° .

The small indentation around the double pole at $s = 0$ must also be mapped.



10.5 Stability via the Nyquist Diagram

Once the complete Nyquist diagram is known:

1. Count the open-loop RHP poles P .
2. Count the net CCW encirclements N of $-1 + j0$.
3. Compute

$$Z = P - N.$$

4. Conclude:

$$Z = 0 \Rightarrow \text{closed-loop stable}.$$

Quick Check

For stability, the required net number of CCW encirclements is

$$N = P.$$

How Open-Loop Instability Changes the Requirement

Open-loop RHP poles	Required N	Stable-case interpretation
$P = 0$	$N = 0$	No net encirclement of -1
$P = 1$	$N = 1$	One CCW encirclement of -1
$P = 2$	$N = 2$	Two CCW encirclements of -1

Important Insight

An open-loop unstable system can be stabilized by feedback, provided the Nyquist plot produces the required encirclements of the critical point.

Worked Example 2: Find the Critical Gain

Consider the unity-feedback loop

$$L(s) = \frac{K}{s(s+3)(s+5)}.$$

For $K = 1$,

$$L(j\omega) = \frac{-8\omega^2 - j\omega(15 - \omega^2)}{64\omega^4 + \omega^2(15 - \omega^2)^2}.$$

The negative-real-axis crossing satisfies

$$\operatorname{Im} L(j\omega) = 0 \quad \Rightarrow \quad \omega^2 = 15.$$

Hence,

$$\boxed{\omega = \sqrt{15} \text{ rad/s}}, \quad \boxed{L(j\sqrt{15}) = -\frac{1}{120}}.$$

Worked Example 2: Stability Range

The open-loop poles are

$$0, -3, -5.$$

The origin is indented, and there are no open-loop poles in the open RHP:

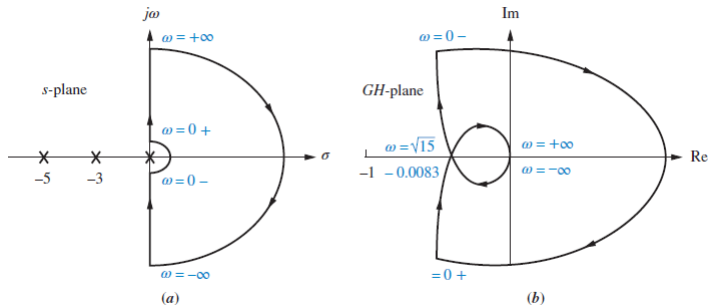
$$P = 0.$$

For marginal stability the scaled real-axis crossing reaches -1 :

$$-\frac{K}{120} = -1 \quad \Rightarrow \quad \boxed{K_{cr} = 120}.$$

Gain	Closed-loop behaviour
$0 < K < 120$	Stable
$K = 120$	Marginally stable, $\omega_{osc} = \sqrt{15}$ rad/s
$K > 120$	
	Unstable

Worked Example 2: Geometric Interpretation



Changing a positive scalar gain K scales the Nyquist locus radially.

$$K = K_{cr}$$

when the locus passes through

$$-1 + j0.$$

At that point the closed loop is marginally stable.

Worked Example 3: Another Gain-Stability Example

Consider

$$L(s) = \frac{K}{(s^2 + 2s + 2)(s + 2)}.$$

All open-loop poles are in the LHP, so

$$P = 0.$$

For $s = j\omega$, the denominator becomes

$$(2 - \omega^2 + j2\omega)(2 + j\omega) = (4 - 4\omega^2) + j\omega(6 - \omega^2).$$

Real-axis crossing:

$$\omega(6 - \omega^2) = 0 \quad \Rightarrow \quad \boxed{\omega = \sqrt{6} \text{ rad/s}}.$$

At this frequency,

$$\boxed{L(j\sqrt{6}) = -\frac{K}{20}}.$$

Worked Example 3: Stability Conclusion

Marginal stability occurs when

$$-\frac{K}{20} = -1,$$

therefore

$$K_{cr} = 20.$$

Gain	Closed-loop behaviour
$0 < K < 20$	Stable
$K = 20$	Marginally stable
$K > 20$	Unstable

At the stability boundary,

$$\omega_{osc} = \sqrt{6} \text{ rad/s}.$$

Nyquist and Imaginary-Axis Pole Crossing

At the critical gain,

$$L(j\omega_c) = -1.$$

Thus,

$$1 + L(j\omega_c) = 0.$$

Therefore the closed-loop characteristic equation has roots at

$$s = \pm j\omega_c.$$

Connection with Root Locus

The Nyquist critical-point crossing and the root-locus imaginary-axis crossing describe the **same stability boundary** from two different viewpoints.

For Worked Example 3,

$$K_{cr} = 20, \quad \omega_c = \sqrt{6} \text{ rad/s}.$$

Nyquist Stability: A Reliable Problem-Solving Procedure

1. Form

$$L(s) = G(s)H(s).$$

2. Count open-loop RHP poles P .

3. Check for poles on the imaginary axis; indent if necessary.

4. Evaluate $L(j\omega)$ for $0 \leq \omega < \infty$.

5. Use symmetry to complete the plot.

6. Mark $-1 + j0$.

7. Count net CCW encirclements N .

8. Compute

$$Z = P - N.$$

9. Stable iff $Z = 0$.

Nyquist Criterion: Key Takeaways

- ▶ Nyquist converts the **open-loop frequency response** into a test of closed-loop stability.

- ▶ The critical point is

$$-1 + j0.$$

- ▶ With the convention used here,

$$Z = P - N.$$

- ▶ Closed-loop stability requires

$$N = P.$$

- ▶ Poles on the imaginary axis require an indentation of the Nyquist contour.
- ▶ Varying loop gain scales the locus and allows the **critical gain** to be found graphically.

Thank You

Questions and Discussion